

Weather Impact on Flights Operations Dashboard

Business

Economy

First

Premium Economy

 **Total Flights**
10000
Weather Impact 1507

 **10.26** Minutes
Weather Delay
Average

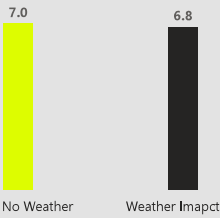
 **77%**
Weather Impact On-
Time Rate

 **83**
Weather Related
Cancellations

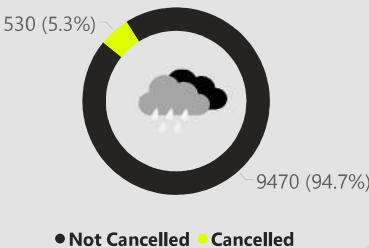
 **6%**
Weather Impact No-
Show Rate



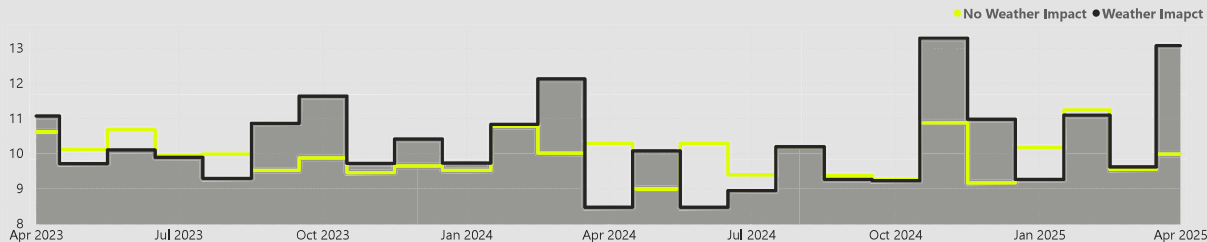
Flight Average Satisfaction by Weather Condition



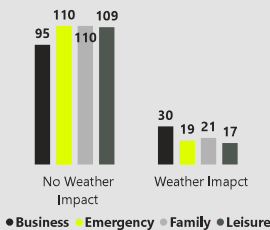
Cancellation in Bad Weather



Flight Delay Over Time by Weather



No Show Count by Weather & Travel Purpose



Weather Impact on Flights Operations Insights

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Executive Summary (Key Takeaways)

- 1.) Out of **10,000 total flights**, only **15% (1,507 flights)** were impacted by adverse weather.
- 2.) Despite weather challenges, the airline maintains a **77% on-time performance**, indicating strong operational resilience.
- 3.) **Weather-related cancellations are minimal (6%)**, showing effective contingency planning.
- 4.) Passenger satisfaction drops slightly during bad weather but remains **above 6.5**, which is still acceptable in aviation benchmarks.

Operational Performance Insights



1.) Weather Impact on Flight Operations

Daily average weather-impacted flights: ~10 flights/day

This suggests weather disruptions are **frequent but manageable**, not extreme or sporadic. Only **83 flights were cancelled due to weather**, which is **less than 1% of total flights**.

Insight:

Weather affects schedules more through **delays than cancellations**, indicating proactive re-routing and recovery strategies.

2.) Flight Delays Trend Analysis

Delays due to weather generally fluctuate between **8–12 minutes**, peaking during certain months. Flights **without weather impact remain stable** with consistently lower delay durations. Seasonal spikes indicate **monsoon/winter impact periods**, which can be forecasted.

Insight:

Weather-related delays are **predictable and seasonal**, allowing scope for **pre-emptive scheduling buffers**.

Customer Experience & Satisfaction

Condition	Avg Satisfaction
No Weather Impact	7.0
Weather Impact	6.8

Insight:

- Even during adverse weather, satisfaction drops only **0.2 points**, showing:
- Effective communication
 - Good handling of disruptions
 - Passenger trust in operations

Strategic Recommendations

Operational Improvements

- Introduce **seasonal buffer time** during high-risk weather months
- Enhance **predictive weather analytics** to reduce delay spikes

Customer Experience

- Improve **real-time passenger alerts** during weather disruptions
- Offer **proactive rebooking/compensation** to maintain satisfaction above 7.0

Business Strategy

- Focus resilience investments on **high-value segments (Business& Emergency travel)**
- Use weather insights to optimize **crew & aircraft allocation**